

ACSD: Authorized Lineage for Anonymous Scholarly Releases

Anonymous authors

Abstract

Anonymous scholarly disclosure needs a more precise answer than “put a signed PDF in a repository.” A signature identifies a signing key, not necessarily a person; a repository timestamp is not independent time; and an exact predecessor hash does not say who may publish the next version. This paper presents ACSD, a narrow executable profile for anonymous scholarly releases and authorized lineage. A compact approval target binds exact release, governance, provenance-policy, and lineage-transition digests before every required pseudonymous key signs it. An unchanged predecessor authority can authorize an exact child through those same approvals; a key-set or threshold change additionally requires a predecessor-quorum set of signatures over the exact transition. The optional series layer remains separate, permitting cyclic semantic citation while exact release commitments stay acyclic. The implementation uses tagged Ed25519/EdDSA COSE, a closed capability calculus, and independently pinned RFC 3161 evidence. Evaluation includes an executable fresh-key $n+1$ attack, authorized rotation and fork cases, 64 fixed and 1,000 generated Python–Node canonical-JSON cases, adversarial timestamp tests, and 21 axiom-free Lean theorems. None of these mechanisms proves natural-person identity, contribution truth, originality, peer review, or legal nonrepudiation.

1 Introduction

Researchers often want to disclose a manuscript before conventional publication while keeping the author identity unavailable during review. A common informal proposal is to upload an anonymous file, sign it, and rely on repository history. That proposal leaves several distinct questions entangled.

One concrete trigger is the interval around an arXiv submission: a first-time or new-category submission may require endorsement, while moderation and review are separate from scientific peer review. During that interval an author may want a public, content-anonymous evidence package that preserves exact bytes and team assent without pretending to replace arXiv’s process. We use this as an application scenario, not as a claim that arXiv queues have a measured backlog or that ACSD should circumvent its rules.

1. What exact bytes did one or more keys approve?
2. Can a later verifier distinguish a revision, a legitimate branch, and an equivocation?
3. Can a sequence of papers cite one another, including mutual citations, without a circular content-hash dependency?
4. What does a time witness establish, and what does it not establish?

The first question is about key control and integrity. It is not, by itself, a proof that a particular natural person wrote the work. The second and third are data-model and verification questions. The fourth requires an external witness such as an RFC 3161 timestamp authority (TSA), not a local clock or a mutable Git field. Conflating these layers produces systems that overclaim.

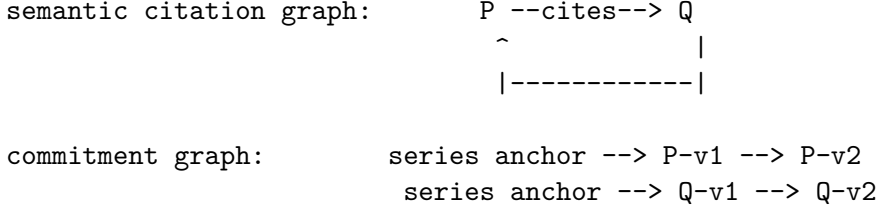


Figure 1: ACSD separates cyclic semantic citations from acyclic exact-byte commitments. Stable random WorkIDs support the first graph; release digests and predecessor links support the second.

This paper presents ACSD (Anonymous Scholarly Claim and Disclosure), a deliberately narrow application profile. It uses standard content hashing, COSE signatures, and provenance concepts rather than proposing a new primitive. The profile is designed for content-anonymous release: the public paper and artifact do not name authors, but an account hosting them can still be linkable. It is therefore *not* described as author-unlinkable double blind publication.

The main contributions are:

1. a self-contained paper-release object with ordered anonymous author slots and per-author endorsements;
2. a two-layer construction that permits cyclic scholarly citations over stable WorkIDs while keeping content commitments acyclic;
3. signed UTF-8 byte-range citation witnesses and a restricted canonical JSON profile to reduce field-interpretation ambiguity;
4. explicit classifications for branches, equivocation, and copied content under a fresh lineage;
5. an acyclic unanimous-approval target and closed capability calculus that prevent unsigned policy amplification and approval reuse after governance or PEC substitution;
6. an exact lineage transition requiring predecessor-quorum signatures, rejecting a fresh-key $n+1$ claimant while supporting explicit author/key changes and branches; and
7. an executable prototype with one-command release and revision paths, independent COSE verification, adversarial RFC 3161 checks, generated differential testing, and an axiom-free Lean model of the composition rules.

2 Scope and Threat Model

Assets. A release contains manuscript bytes, an ordered list of anonymous author keys, roles and contribution declarations, an AI-use declaration, and optional parent-release information. A series package contains references to completed releases, a series key, sequencing information, and cross-paper semantic relations. Public keys are pseudonymous identifiers.

Adversary. We consider a party that can copy public content, generate fresh keys, reuse the exact WorkID and predecessor digests, claim the next unused version number, reorder or substitute unsigned metadata, present a signed but malformed JSON payload, and claim unsupported cross-paper citations. We also consider a series key that signs two incompatible objects. We do not

assume that a signature reveals a human identity, that a contribution declaration is factually true, or that a local timestamp is trusted.

Non-goals. ACSD does not prove authorship by a natural person, independent discovery, plagiarism, venue acceptance, peer review, or legal nonrepudiation. These are not merely unimplemented features: some require social, institutional, or legal evidence outside the cryptographic object. The profile instead preserves evidence that can be useful in a later dispute.

3 Release Layer

3.1 Canonical release object

Each *PaperRelease* has a random stable *WorkID*, a visible slot (*WorkID*, *version*, *line*), a content path, content length and digest, ordered author records, declared roles, contributions, an AI-use declaration, optional exact predecessor, and a list of citation witnesses. The release identifier is

$$\text{ReleaseID} = \text{urn:sha256:}H(\text{canonical}(R)).$$

The content digest is over exact manuscript bytes. The release object uses a deliberately restricted JSON encoding: recursively sorted ASCII object keys, no insignificant whitespace, and safe integers only. A verifier parses, re-canonicalizes, and compares bytes before accepting a signature. This is a profile rule, not a claim that it is a general JSON standard.

3.2 Joint approval target and closed capability calculus

The *AuthorshipGovernanceStatement* is the exact declaration of the work id, manuscript digest, ordered byline slots and roles, corresponding-author key, and AI-use field. The CLI derives it from the team configuration and release; there is no privileged coordinator whose unsigned word makes it authoritative. Instead, every declared author key approves its digest through the common target described below. This proves assent to the declaration, not the factual truth of its roles or contribution history.

A *Provenance Evidence Capsule* (PEC) binds the release subject and governance digest to a digest-linked evidence-event sequence and a claim policy. The policy contains a fixed-vocabulary list of permitted outcomes, their required capabilities, and mandatory non-claims. A PEC also names one team key as an issuer for provenance, but that label alone grants no outcome: the complete team must still approve the common target.

The v3 command-line profile constructs an acyclic `acsd-approval-target/v2` object. It points to already complete objects and is itself signed only through detached approval envelopes. The following is a structurally valid, pretty-printed example with illustrative digest values:

[illegible]

Every required key issues a tagged COSE Sign1 signature over the same canonical target bytes. Verification recomputes all four bindings and each key id from the supplied public key. The closed capability calculus is the rule

$$\text{Granted}(P, o) \iff \text{Accepted}(P) \wedge o \in P.\text{permittedOutcomes} \wedge \text{CapabilityEstablished}(P, o).$$

Policy permission is therefore only an upper bound; it is not evidence. The outcome type is closed to four machine conclusions, so an unknown social claim cannot enter the rule by adding a string to JSON.

Without the joint target, suppose authors sign only the release while an attacker later changes the unsigned governance object or PEC policy. The attacker could relabel roles, add the external-time outcome, or try to map a local clock to that outcome while the old release signatures still verify. Under ACS D, changing the governance or policy changes its digest and causes a target-binding failure; recomputing the target instead causes the old COSE approvals to fail their payload check. Even an approved policy entry yields no outcome without its separately verified capability. A verifier can thus state that the pseudonymous keys jointly approved this exact tuple, but cannot infer who controls them without separate enrollment evidence.

3.3 Authorized succession rather than version-number possession

An exact predecessor digest proves structural continuity, not authority. A fresh-key attacker can copy a public WorkID, name the real predecessor, choose the consecutive version number, and validly sign the resulting child with the attacker’s keys. No hash or slot-conflict check answers whether those keys were entitled to create the parent-to-child edge. ACS D v3 therefore binds every release to a *lineage authority*: a sorted author-key set and a threshold for authorizing the next edge. The default threshold is unanimity. It is stored in the predecessor, so a child cannot lower it unilaterally.

Let P be an accepted predecessor, C a proposed child, A_P the authority bound by P , and A_C the authority proposed by C . The verifier accepts a lineage edge only if

$$\text{AuthorizedSuccessor}(P, C) = \text{ExactEdge}(P, C) \wedge \text{Authorization}(P, C),$$

where

$$\text{Authorization}(P, C) = \begin{cases} \text{Quorum}(A_P, \text{ApproveTarget}(C)), & A_P = A_C, \\ \text{Quorum}(A_P, \text{Transition}(P, C)) \wedge \text{AllApprove}(A_C, C), & A_P \neq A_C. \end{cases}$$

Here **ExactEdge** requires the same WorkID, exact parent release and PEC digests, and either a same-line increment by one or a new named line beginning at version one. **Transition** binds both authorities and the exact child release, governance, and PEC digests. The child approval target binds the transition digest. The construction is acyclic because the transition refers to complete child bodies while detached approvals refer to their completed digests.

For an unchanged authority object, the ordinary child approvals already constitute predecessor authorization and no duplicate signing step is needed. A key-set or threshold change uses double control: the predecessor threshold signs the exact transition and every new author signs the child target. This resembles the old-and-new authorization pattern used for TUF root-key updates, but ACS D applies it to a single scholarly lineage edge rather than software repository metadata [8]. A series key may advertise a verified child as a head; it cannot manufacture predecessor-author assent by itself.

Consider the concrete $n+1$ attack. Mallory constructs a child with P's WorkID, P's exact release and PEC digests, the next version number, Mallory's public key, and a valid Mallory signature over every child-controlled object. The object is internally signed but has neither an unchanged predecessor authority nor a transition signed by A_P . The implementation returns `VALID_OBJECT_BUT_UNAUTHORIZED_SUCCESSOR`. It does not accuse Mallory of plagiarism; it rejects only the claimed lineage edge.

This conclusion is relative to an accepted exact predecessor, not to the WorkID string alone. The CLI can pin a previously accepted parent with `-expected-parent-release-id`; otherwise it reports `UNPINNED_EXACT_PARENT`. Without an external parent anchor, an isolated package proves only that the authority named by its embedded parent authorized the edge; it cannot distinguish a separately fabricated genesis that copied a public WorkID.

Threshold choice exposes a real safety–liveness tradeoff. Unanimity prevents one coauthor from transferring the lineage alone but lets one lost or dissenting key freeze it. A lower threshold improves availability and is therefore allowed only when committed by the predecessor. There is no post-hoc identity recovery in this profile: without a precommitted recovery authority, lost threshold keys safely stop the cryptographic lineage. Nor can a later message erase a valid past authorization. Two valid distinct children in the same exclusive slot are reported as equivocation with no automatic winner; choosing by observation time requires an external transparency or governance policy.

3.4 Citation witnesses

The prototype recognizes only literal occurrences of a canonical token `urn:uuid:<work-id>` in the UTF-8 manuscript bytes. For every occurrence it records and signs

(`WorkID`, `byteOffset`, `byteLength`).

The published `reference_work_ids` list must equal the ordered de-duplicated projection of these witnesses. This prevents a series package from inventing a citation edge that has no corresponding token in its source release, and prevents a copied text from silently relabeling an old target as a new WorkID.

The rule intentionally does not parse PDFs, infer references from natural language, or determine whether a citation is intellectually appropriate. It is an exact byte-binding contract, not a semantic understanding claim.

For example, one UTF-8 source line contains the following token:

```
urn:uuid:12345678-1234-1234-1234-123456789abc
```

The witness records that `WorkID`, the byte offset of the first character of the token, and its byte length (45 bytes for this token). An editor that inserts a character before the token must regenerate the witness. This is intentionally mechanical: a source-level helper or build macro can insert the token, but v1 does not yet provide such editor integration, and hiding the token in a PDF would defeat this scanner.

A minimal release fragment is conceptually:

```
{
  "slot": {
    "line": "main",
    "version": 1,
    "work_id": "urn:uuid:..."
  },
  "content": {
    "sha256": "...",
    "reference_work_ids": ["urn:uuid:..."],
    "citation_witnesses": [
      {
        "work_id": "urn:uuid:...",
        "byte_offset": 35,
        "byte_length": 45
      }
    ]
  }
}
```

The ellipses above are presentation placeholders only. The released demo fixture binds a real manuscript’s bytes, digest, and work identifier (the governance and PEC objects are real, not placeholders), using synthetic author identifiers. Real series releases are produced by the v1 fixture generator.

4 Series Layer and Cyclic Citations

4.1 Two graphs, two purposes

A paper remains verifiable after all series packages are removed. The optional series layer records relationships among independently complete releases. A signed series package contains

$$B_k = \text{Sign}_S(\text{seriesId}, \text{epoch}, \text{sequence}, \text{previousBundleHash}, \\ \text{rotationHash}, \text{rank}, \text{releases}, \text{heads}, \text{semanticEdges}).$$

The semantic graph has vertices `WorkIDs` and may contain $P \rightarrow Q$ and $Q \rightarrow P$. Its edges express declared citation or related-work relations. The commitment graph has series anchors and completed `ReleaseIDs`. It does not make a release depend on a future series package. Thus two authors can cite stable random `WorkIDs` before both final documents are content-addressed; a later package resolves each `WorkID` to a completed `ReleaseID`. The semantic graph may be cyclic while the commitment graph remains a DAG.

The package includes a signed structural rank strictly above every release version it resolves. Combining anchor-to-release edges with exact release-to-parent edges decreases this rank along a commitment path. This rank is a graph certificate, not a timestamp, publication date, or priority claim.

The package fields have narrow roles: *epoch* names the active series key generation; *rotationHash* commits its authorized key transition; *heads* maps each `WorkID` to its advertised line head; and *semanticEdges* records declared citation or related-work relations. The rank is only a finite measure certifying that the commitment graph has no cycle.

4.2 Revisions, branches, and conflicts

A revision names one exact predecessor. Multiple children of a parent are allowed, because a main manuscript and a computational appendix can be legitimate independent lines. A paper-level equivocation is instead detected when two valid, different releases occupy the same exclusive slot. A series-level equivocation is detected when two valid packages share the same series, epoch, sequence, and predecessor hash while committing to different contents.

An adversary can copy the content bytes, issue a fresh `WorkID` and keys, and obtain a valid new lineage. ACSD classifies this as content reuse under a different lineage, not as a cryptographic proof of plagiarism. That restraint is important: reuse may be authorized, collaborative, or independently derived.

As a concrete attack, suppose an attacker copies P-v1, changes its `WorkID`, and asks the series package to claim that it cites Q. The release verifier recomputes the content digest and byte witnesses; if the copied source does not contain the declared Q token, the package is rejected as an unsupported edge. If the attacker copies the token too, the package can still be valid under a new lineage. The evidence then says only “content reuse under a different lineage”; deciding plagiarism or authorization remains outside the verifier.

5 Time Evidence

Time is intentionally external to author signing. An RFC 3161 request contains the SHA-256 imprint of the frozen approval target. A response supports the limited statement that the TSA saw that imprint no later than its recorded `genTime` only when the verifier supplies a TSA signer-certificate or fingerprint pin independently of the package. A certificate and its fingerprint copied into the same mutable package are not a trust anchor. The adapter checks the request/response nonce, SHA-256 imprint, `id-ct-TSTInfo` content type, single signer identifier, critical and exclusive timeStamping EKU, ESS certificate identifier, signed attributes, and CMS signature. It implements an exact-signer pin, not general PKIX path construction or revocation. Its local test TSA is a protocol test and can never produce `EXTERNALLY_NOT_AFTER`.

There is no circularity: the timestamp receipt is a sidecar over the already signed target. A later revision can bind the receipt hash if a new signed object must refer to it. A timestamp does not prove creation at that moment, authorship, originality, or global first discovery. Multiple independently governed services can reduce single-service reliance, but their composition requires a separate policy.

6 Implementation and Evaluation

The v1 prototype uses Python for deterministic fixture generation and a zero-dependency Node.js verifier for strict CBOR/COSE and profile validation; the two paths are intentionally separate. The package is designed for offline replay: locked dependencies are included, and the verification driver regenerates fixtures, verifies a standalone release, independently verifies the series profile, builds a manifest, and verifies that manifest.

A command-line tool (`acsd.py`) turns the profile into an executable workflow: `keygen`, `init`, `authorize`, `approve`, `finalize`, `verify`, and `compare-successors`, plus one-command `release` and `revise` paths. Approval status is derived from verified COSE files, never trusted from mutable state metadata. COSE uses the registered EdDSA value `-8`, requires tag 18, and rejects non-minimal or ambiguous CBOR encodings. A zero-dependency Node path independently verifies Python-produced approvals. The canonical JSON layer is exercised by 64 fixed Python–Node vectors plus 1,000 seeded generated cases; all generated cases agree on canonical bytes and text-layer verdicts. Three known object-layer float typing differences do not change the text-layer rejection result and remain recorded in the canonical specification.

The negative corpus includes the inherited rank, unsupported-edge, citation-offset, canonical-JSON, and path attacks. New cross-layer regressions replace a public key under an old key id; coherently replace governance and PEC; recompute an unsigned target and attempt to reuse old approvals; alter COSE tags, algorithm identifiers, map keys, and lengths; replace a timestamp and rewrite the transport manifest; mismatch the nonce, signer, EKU, TSTInfo content type, or signature algorithm; and attempt to promote the local test TSA. All are rejected at their intended boundary. The single freeTSA result is an existence check that the adapter can consume one real service response, not a reliability study or a multi-provider conformance survey.

The lineage corpus begins with the smallest discriminating attack: a child with the exact WorkID and parent release/PEC digests, `n+1`, a fresh attacker key, and complete self-approval. It is rejected at the missing predecessor-quorum boundary. Positive cases cover unchanged authority, key rotation, one-of-two and two-of-two thresholds, and a named branch; negative cases cover threshold downgrade and transition-signature replay. Two valid same-slot siblings yield equivocation with no winner.

Table 1: Executable evaluation corpus. The first seven rows are inherited v1 fixture results; the remaining rows are reproducible from v3.

Object or check	Count / result
Standalone release versions	8 / 8 valid without series packages
Author endorsements	18 / 18 bound to the intended release
Signed series objects	7
Closed normal, attack, and indeterminate scenarios	13 / 13 expected outcomes
Independent profile checks	30 / 30 passed
Canonical JSON regression cases	11 / 11 Python–Node agreement
Public-manifest entries	113 / 113 verified
v3 differential canonical-JSON vectors	64 / 64 accept/reject agreement
Offline v3 unit-test methods	59 passed; 1 network test skipped
Seeded generated canonical cases	1000 / 1000 byte and verdict agreement
Independent approval implementations	Python producer / Node verifier
Authorized-lineage scenarios	8 / 8 expected outcomes
Real-TSA existence check	1 freeTSA response verified
PEC and lineage theorems in the abstract core	21; no sorry/admit

The measurements remain prototype-scale rather than a throughput claim. They were recorded under Python 3.12.14 on 64-bit Windows 10.0.26200, a 13th Gen Intel Core i7-1360P with 16 logical processors, and 31.7 GiB physical memory. In that run, five repetitions of an in-memory workload that canonicalizes objects, constructs and verifies SHA-256 manifest entries, and scans a linear rank certificate had median times of 2.23, 26.45, 227.54, and 718.81 ms for 10, 100, 1,000, and 5,000 objects respectively; peak traced memory at 5,000 objects was 7.04 MiB. This excludes filesystem, COSE, and network cost, so it is a smoke test for obvious scaling anomalies, not a statistically powered performance evaluation; dispersion beyond minimum and median was not retained.

The v3 Lean development uses concrete approval targets, approvals, digest-linked events, a closed outcome type, decomposed time evidence, and typed lineage authorities and transitions. Its 21 theorems prove, among other facts, that an approval for one target digest cannot be reused for a different target; a granted result implies acceptance and policy permission; key assent requires all target approvals; and external time requires exact-subject, signer-pin, nonce, profile, and non-local predicates. The lineage theorems show that every accepted successor has predecessor-quorum evidence, a fresh authority cannot enter through the unchanged-authority path, and a transition bound to one child digest cannot authorize another child. The model deliberately accepts signature validity and digest equality as inputs. It does not prove parser refinement, SHA-256 collision resistance, Ed25519 unforgeability, TSA honesty, or general PKIX correctness.

These 21 are PEC, lineage, and cross-layer composition theorems, not a recount of the separately published v1 Lean core.¹ The v1 artifact records 53 theorem statements about release catalogs, teams, lineage, series graphs, citation witnesses, and lifecycle rules. That earlier model remains an inherited, distinct proof surface; it was neither deleted nor compressed into the new count. The v3 count concerns the narrower executable model of approval-target reuse, capability gating, event links, external-time preconditions, and authorized succession.

¹The v1 proof surface is archived at <https://github.com/huiminZheng-collab/acsd-anonymous-series-v1/tree/main/formal-core>.

7 Related Work

The profile builds on established foundations. PROV-DM supplies a general vocabulary for entities, activities, agents, derivations, and bundles [1]. Software Heritage identifiers provide intrinsic identifiers for software objects [2]; Trusty URIs provide hash-bearing immutable artifact identifiers [3]. COSE defines the signed binary envelope used for endorsements [4]. SCITT specifies a broader architecture for signed statements and transparency receipts [5]. RFC 3161 specifies the timestamp protocol [6]. TUF uses predecessor and successor root thresholds to preserve trust through root-key updates [8]; in-toto binds supply-chain steps to authorized functionaries [9]. These are the closest authorization patterns, although neither specifies anonymous scholarly WorkID succession.

ACSD does not claim to replace these systems or invent threshold delegation. Its narrow contribution is an executable scholarly profile that combines: self-contained anonymous paper releases; exact predecessor-authorized succession; a separate series layer for mutual citations; exact text-token witnesses; explicit conflict predicates; and counterexamples against common overclaims about signatures, local clocks, and copied content.

Table 2: Capability boundary (not a performance comparison).

System	Primary capability	ACSD boundary
OpenTimestamps	Blockchain-anchored timestamp proofs	Optional time sidecar; no author-role or citation semantics
Software Heritage / SWHID	Persistent intrinsic identifiers for archived software objects	ACSD binds scholarly release roles, declarations and paper relations
Zenodo	Versioned archival record and DOI	ACSD’s byte-level policy and pseudonymous team assent remain explicit
RFC 3161	TSA-signed message-imprint time evidence	Exact not-after evidence only; never authorship or originality
TUF / in-toto	Threshold key update and authorized software-supply-chain functionaries	ACSD scopes old/new authority to exact scholarly child objects
ACSD	Release, governance, authorized revision, witness and claim-policy profile	Composes these boundaries while retaining backend trust assumptions

8 Limitations and Future Work

Several extensions are intentionally deferred. First, relations across independently administered series require policy and an interoperable cross-series package model. Second, anti-equivocation evidence from transparency logs or gossip requires deployed service assumptions. Third, the parser, threshold counter, canonicalization, hash, and signature implementations have not been refined into the Lean model. Fourth, generated private keys are unencrypted PEM files; a production CLI needs encrypted or operating-system-backed storage. Fifth, exact TSA signer pinning does not replace PKIX path construction and revocation when a deployment requires those services. Sixth, v3 has no precommitted recovery authority: loss of enough predecessor keys freezes the lineage, while an already valid transfer cannot be retroactively revoked. Finally, content-anonymous publication does not hide repository-account, network, timing, or public-key linkage, and richer citation/editor

integration remains future work.

These are reasons to state the present result narrowly, not reasons to weaken the useful evidence it already preserves.

9 Conclusion

ACSD turns a vague “anonymous signed paper” idea into a limited but executable evidence profile. It binds exact released objects to pseudonymous-key approvals, distinguishes a signed object from a predecessor-authorized successor, supports ordered roles and branches, permits mutual scholarly citation without cyclic hash dependencies, and detects same-slot authorized forks without inventing a winner. It leaves human identity, originality, recovery without prior authority, and independent time as claims requiring additional evidence. This division of claims is the central design principle: preserve what cryptography can verify, and make the remaining social questions explicit.

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